

Persistent Post-Concussive Symptoms (PPCS) — Clinical Flowchart

PPCS = symptoms persisting > 4 wks adolescents / > 10–14 days adults · 10–20% of concussion patients · requires re-assessment, phenotype ID, and targeted rehab

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Confirm PPCS and exclude red flags

Re-take history. Any change since last review? Any new injury event? Any new symptom development?

At ANY point in the PPCS pathway, escalate if: new focal neuro deficit · GCS ≤ 13 or declining · seizure · severe / worsening headache · repeated vomiting · new anticoagulant + head symptom in older patients



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SCOAT6 re-administration + systematic screen

Longitudinal anchor — compare to previous scores. Screen systematically across domains:

- Symptom inventory (0–132)
- Cognitive screen
- Cervical exam + cervico-neurological
- VOMS
- Gait + balance (tandem)
- Orthostatic HR
- Mood — PHQ-9, GAD-7
- Sleep — ISI



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Identify dominant phenotype(s) → guides rehab selection

Most PPCS patients have 2–3 coexisting phenotypes. Target the most limiting first.

Phenotype	Screens positive if...	First-line intervention
Vestibular	Abnormal VOMS, motion sensitivity, positional dizziness	Vestibular rehab ± BPPV repositioning
Oculomotor	NPC > 5 cm, abnormal saccades/pursuit, screen intolerance	Vision therapy ± neuro-optometry referral
Cervicogenic	Cervical exam findings, positive head-neck differentiation	Cervical manual therapy + deep neck flexor
Migrainous	Pulsatile headache, photo/phonophobia, prior migraine	GP review for pharmacological management
Affective	PHQ-9 ≥ 10 or GAD-7 ≥ 10	Psychology (CBT); GP if severe
Cognitive	Disproportionate cognitive sx, exertion-gated fatigue	Graded cognitive exertion; neuropsych referral
Sleep	ISI > 14, disrupted architecture	Sleep hygiene + CBT-I
Autonomic	Orthostatic intolerance, POTS features	BCTT-guided aerobic + GP for POTS workup



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Phenotype-directed active rehabilitation

Sequencing matters. Treat the most limiting cluster first. Cervical-before-vestibular sequencing produces better outcomes than parallel (Reneker 2021).

- Active rehab, not passive rest
- Sub-symptom threshold progression (BCTT-guided)
- Address mood, sleep, pain concurrently
- Realistic goal-setting — measured in weeks to months
- Fortnightly review with symptom tracking



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Multidisciplinary team referral

TIER 1 — always

GP · EP · Psychology

TIER 2 — phenotype-specific

Sports physician · Neuro-optometrist · OT ·
Neuropsychologist · Physio/osteo/chiro

TIER 3 — specialist

Neurologist · Endocrinologist · Psychiatrist ·
Pain specialist



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Fortnightly review + treatment-failure check

At each review: SCOAT6 · phenotype re-assessment · adjust intensity · confirm RTP/RTL/RTW progression

Escalate / reconsider differential if: symptoms worsen despite sub-threshold approach · no improvement over 4 wks targeted rehab · new symptom clusters · fear-avoidance developing · exacerbation not explained by current framework